

Design Note: MIND & EB-NeRD News Recommender

Built one retrieval pipeline (unified schema, native temporal splits, leakage-guarded features, a shared BM25/semantic eval harness) that runs against two datasets, MIND (English, 104,151 train+val articles / 376,471 val impressions / 120,961 test articles, 2.37M test impressions) and EB-NeRD (Danish, `ebnerd_small`, 20,738 articles / 244,647 val impressions). Three strategies were built in sequence on MIND; EB-NeRD's build tracks Strategy B.

1. Strategy A: Plain BM25 + Plain LSA

Built:

- **BM25:** Unweighted single-field BM25 using `CountVectorizer` → sparse CSR weight matrix.
- **LSA:** TF-IDF + `TruncatedSVD`, using 128 dimensions with no entity fusion.
- **Evaluation:** Both methods were scored using the two scoring approaches described above.
- **Offline harness:** Evaluated AUC, MRR, nDCG@5, nDCG@10, diversity, novelty, and coverage with bootstrap confidence intervals.

Key choices:

- **Native-file splits:** MIND's/EB-NeRD's own train/val/test files are already temporally disjoint by construction; asserted monotonic rather than re-derived from a day cutoff.
- **LSA for the first semantic pass:** Used LSA instead of a transformer to avoid new dependencies. It fits ~100K articles in single-digit seconds, keeping the rest of the pipeline (schema, leakage guards, harness) iterable.
- **Separate scoring APIs:** Used two scoring paths from day one: full-corpus scoring and restricted-candidate scoring, rather than a single generalized function.

Alternatives considered:

- **rank_bm25:** Rejected because it scores one query/document pair at a time in Python, has no vectorized batch path, and lacks the field-weighting support needed from Strategy B onward.
- **Elasticsearch/Solr:** Rejected because this is an offline batch-scoring job rather than a live-serving system. A search cluster would add operational overhead without any concurrency requirement.

Observations:

- **Lexical/semantic divergence:** Appears immediately on both datasets, even when unweighted and untuned.
- **Reranking:** LSA narrowly outperforms BM25 in MIND reranking AUC (0.5591 vs. 0.5581).
- **Full-corpus recall:** BM25 wins by a wide margin (recall@50: 0.551% vs. 0.307%). This is a structural property of exact-ID retrieval versus small-list reranking, rather than an effect introduced by later tuning.
- **MIND leaderboard:** The score at this stage was 0.5805.
- **EB-NeRD:** Its pipeline was built directly on a more advanced configuration (field-weighted BM25, tuned k_1/b , and sentence-transformer). No separately measured plain/untuned EB-NeRD baseline exists, due to prioritization given its much smaller corpus and lower iteration headroom.

Breaks at 10×

- **No persisted index:** Every script (retrieval evaluation, ranking evaluation, and prediction generation) re-fits BM25+LSA from scratch on every invocation.
- **Repeated fitting cost:** At 10× articles (~1M), the fitting cost is paid three separate times per experiment instead of once.
- **Dense full-corpus scoring:** Full-corpus scoring uses a dense `.todense()` matrix multiplication with complexity $O(\text{batch} \times n_{\text{docs}})$.
- **Compute blow-up:** At 10× articles and 10× impressions simultaneously, the recall@K diagnostic experiences a ~100× compute increase. BM25 alone already takes ~40 minutes at 1× scale.
- **Primary breakage point:** This represents the single sharpest and most predictable scalability limitation in the pipeline.

70	des_007	2026-08-25 04:07	900327	0.5805	0.2799	0.2967	0.3523
----	---------	------------------	--------	--------	--------	--------	--------

Figure 1: MIND Leaderboard score for Strategy A

2. Strategy B: Field-weighted BM25 + tuned k_1/b + entity-fused sentence-transformer / FAISS

Built:

- **Index persistence:** `build_indices.py` fits BM25 and semantic models once, pickles both with a `config.json` sidecar, allowing all downstream scripts to share one fitted index.
- **k_1/b tuning:** Added a grid-search script using restricted-candidate scoring on a fixed subsample.
- **Field-weighted BM25:** Upgraded BM25 with separate title/abstract length normalization and field weights.
- **Sentence-transformer:** Replaced LSA with a pretrained sentence-transformer: `all-MiniLM-L6-v2` for MIND and a multilingual model for EB-NeRD's Danish text.
- **FAISS retrieval:** Added FAISS `IndexFlatIP` for semantic retrieval.
- **Entity fusion:** For MIND, TransE entity embeddings are mean-pooled and concatenated with the content embedding using L2-normalize $\rightarrow$ concatenate $\rightarrow$ renormalize. EB-NeRD has no entity embeddings, so this is MIND-only.

Key choices:

- **Plain pickle:** Used `pickle` instead of a vector database because the pipeline is an offline, rebuild-per-run batch job with no concurrent-query or incremental-update requirements.
- **Custom serialization:** `__getstate__`/`__setstate__` excludes the unpicklable transformer and FAISS index, rebuilding FAISS cheaply from the stored embedding array when loaded.
- **Exact FAISS search:** Used `IndexFlatIP` rather than an approximate index. At 100–120K vectors, exact search is not the measured bottleneck; BM25's full-corpus path is.
- **k_1/b tuning:** Used restricted-candidate scoring on a subsample rather than a full-corpus sweep, making each grid point $\sim 15.8\times$ cheaper for the metric that actually matters.

Alternatives considered:

- **all-mpnet-base-v2:** Rejected because its 768-dimensional embeddings have higher CPU encoding cost. It remains a possible option if quality is later prioritized over latency.
- **Hosted embedding APIs:** OpenAI/Cohere were rejected because per-call network latency is unsuitable for the required impressions/sec, while also introducing a hard third-party dependency into an otherwise offline pipeline.
- **IndexIVFFlat/HNSW:** Rejected at the current scale and planned for reconsideration at larger corpus sizes.

Observations:

- **Largest measured lever:** The semantic-backend swap produced the largest measured improvement in the project. MIND reranking AUC increased from 0.5591 to 0.5860 on the semantic side, bundled with BM25F/tuning work in one commit.
- **MIND leaderboard:** The leaderboard score reached 0.5872 at this stage.
- **EB-NeRD divergence:** BM25 wins full-corpus recall@50 (0.676% vs. 0.438% for the sentence-transformer), while the sentence-transformer wins reranking AUC (0.5366 vs. 0.4949). This confirms the lexical/semantic divergence is dataset-independent.
- **EB-NeRD BM25 anomaly:** BM25 reranking AUC (0.4949) remains *below chance* at every tuned k_1/b tried, despite normal full-corpus recall.
- **Leading hypothesis:** EB-NeRD's front-page logs may be curated for freshness/diversity in a way that works against pure similarity ranking. This was never conclusively confirmed.
- **Cold-start difference:** EB-NeRD's validation split has zero cold-start impressions compared with $\sim 14\%$ on MIND, indicating a dataset-construction difference rather than a pipeline bug.

Breaks at $10\times$:

- **Sentence-transformer encoding:** Encoding is CPU-bound and roughly linear in corpus size. At $10\times$ articles ($\sim 1\text{M}$), a build step that takes “on the order of minutes” at 100K articles could plausibly grow to hours on one CPU.
- **New dominant cost:** Once index persistence removes the redundant-refit problem, sentence-transformer encoding becomes the dominant one-time build cost.
- **FAISS memory:** `IndexFlatIP` requires $O(n)$ memory. At $10\times$ articles, the current embedding + index memory of $\sim 200\text{--}470\text{MB}$ scales to roughly 2–4GB for one semantic index.
- **Approximate search threshold:** This memory growth marks the point where an approximate index, rejected at $1\times$ scale, stops being premature.

69	des_007	2026-08-25 20:01	901506	0.5872	0.2832	0.2999	0.3551
----	-------------------------	------------------	--------	---------------	--------	--------	--------

Figure 2: MIND Leaderboard score for Strategy B

ID # ▾	File name	Date	Status	Score	Detailed Results	Actions
901708	predictions.zip	2026-08-25 23:30	Submitted			⊖

Figure 3: EB-NeRD Leaderboard score for Strategy B (not yet scored)

3. Strategy C: Entity-overlap boost + score fusion + corrected MRR + sparse top- k + multiprocessing

Built (MIND only, see EB-NeRD status below):

- **Entity-overlap boost:** Added a lexical entity-overlap boost directly to BM25 using raw entity-ID/surface-string overlap between query and candidate, *not* embedding-based.
- **Score-level fusion:** Applied per-impression min-max normalization to each scorer’s candidate scores, followed by an α -weighted average with $\alpha = 0.7$ toward the semantic score.
- **MRR correction:** Corrected MRR to average reciprocal rank over *every* clicked candidate in an impression, matching the reference scoring convention instead of considering only the best-ranked click.
- **Sparse BM25 scoring:** Rewrote BM25’s full-corpus path to consume the sparse matrix multiplication row-by-row instead of densifying.
- **Parallel prediction generation:** Parallelized prediction generation across worker processes, with each worker receiving the already-fitted index once at pool startup.

Key choices:

- **Lexical entity boost:** Used ID/string overlap rather than embedding-based similarity. This works even without entity embeddings and could therefore extend to EB-NeRD.
- **Score-level fusion:** Used post-hoc normalization and weighted averaging rather than a learned combiner, keeping the pipeline unsupervised and deferring supervised reranking.
- **Multiprocessing:** Used a pool `initializer` to provide the already-fitted index once per worker rather than once per task, since fitting and pickling the index is the expensive operation.

Alternatives considered:

- **Learned/supervised reranker:** A reranker using features such as (bm25_score, semantic_score, popularity, ...) was identified as the highest-ceiling remaining lever but repeatedly deferred to keep this phase unsupervised.
- **Re-widening the k_1/b grid:** The current sweep reached its boundary in both k_1 and b without plateauing. Despite this, the edge value of the narrow default grid was accepted directly.

Observations:

- **MRR correction:** On the identical sentence-transformer index, with no other change, reported MRR dropped from 0.3188 to 0.2782. BM25’s MRR similarly dropped from 0.3005 to 0.2625.
- **Metric-computation fix:** The similar MRR changes occurred despite only a +0.0004 AUC change from the entity boost, providing strong evidence that the change was a metric-computation fix rather than a regression.
- **Fusion gain:** Fusion produced a small but consistent gain over the semantic backend alone. MIND AUC increased by +0.0040, with every accuracy metric moving in the same direction.
- **Best leaderboard score:** Fusion produced the current best leaderboard score of **0.5953**, although its improvement was smaller than the semantic-backend swap.
- **Diversity and coverage:** The lexical method is consistently the most diverse and provides the best catalog coverage. The semantic method is the least diverse, while fusion inherits a blend of the two.
- **EB-NeRD status:** This strategy was never applied to EB-NeRD. Its build predates this work, and EB-NeRD has no entity embeddings for the fusion component. Therefore, EB-NeRD’s reported numbers remain Strategy-B-equivalent.
- **Known gap:** Backporting the lexical entity boost, fusion, and MRR fix to EB-NeRD is a known and tracked gap, not an oversight.

Breaks at 10×

- **Per-worker index copies:** Each worker receives a pickled copy of the index at pool startup, causing memory usage to scale with $n_{\text{workers}} \times \text{index size}$ rather than just the index size.
- **Memory limitation:** At $10\times$ corpus scale, several workers holding full multi-GB index copies could exceed a single machine’s RAM.
- **Primary new breakage point:** This worker-level memory duplication is the sharpest new scalability limitation introduced by the multiprocessing optimization.
- **Future improvement:** A shared or memory-mapped index could avoid one full index copy per worker. This has not yet been implemented.
- **Sparse top- k rewrite:** The BM25 full-corpus sparse top- k rewrite was verified for correctness but was not re-benchmarked for wall-clock speedup.
- **Scaling caution:** At $10\times$ scale, the runtime improvement must be measured before relying on extrapolated performance estimates.

64	des_007	2026-08-26 03:54	901877	0.5953	0.2884	0.3065	0.3618
----	-------------------------	------------------	--------	---------------	--------	--------	--------

Figure 4: MIND Leaderboard score for Strategy C